

Zabbix Implementation Guide

Step-by-Step Deployment for SecureBank SOC

SecureBank SOC Lab

SIEM Server	192.168.44.10 (Ubuntu 20.04+)
WEB Server	192.168.44.129 (Ubuntu 18.04)
Zabbix Version	6.0 LTS
Date	May 20, 2026

Practical implementation guide with exact commands to deploy Zabbix monitoring infrastructure

Quick Navigation

1. Phase 1: Zabbix Server on SIEM (192.168.44.10)
2. Phase 2: Zabbix Agent on WEB (192.168.44.129)
3. Phase 3: Configure Host in Zabbix Dashboard
4. Phase 4: Verify Monitoring
5. Phase 5: Wazuh File Integrity Monitoring
6. Troubleshooting Reference

Phase 1: Install Zabbix Server on SIEM

Host: SIEM (192.168.44.10) | User: siem

Step 1.1: Verify Internet Connectivity

Action

```
siem@ubuntu:~$ ping -c 3 google.com
siem@ubuntu:~$ ip addr show
```

Expected Result

IP should be `192.168.44.10` , DNS should resolve `google.com` , and ping should succeed with 0% packet loss.

Step 1.2: Install Zabbix Server Packages

Action

```
siem@ubuntu:~$ sudo apt update
siem@ubuntu:~$ sudo apt install -y zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zab
```

If Installation Fails (DNS/Network Issues)

```
siem@ubuntu:~$ echo "nameserver 8.8.8.8" | sudo tee /etc/resolv.conf
siem@ubuntu:~$ sudo apt update
siem@ubuntu:~$ sudo apt install -y --fix-missing zabbix-server-mysql zabbix-frontend-php zabbix-a
```

Step 1.3: Create Zabbix MySQL Database

Action

```
siem@ubuntu:~$ sudo mysql -uroot -p
# Enter your MySQL root password when prompted
```

Then run these SQL commands inside MySQL:

```
CREATE DATABASE zabbix CHARACTER SET utf8mb4 COLLATE utf8mb4_bin;
CREATE USER 'zabbix'@'localhost' IDENTIFIED BY 'SecureBank123!';
GRANT ALL PRIVILEGES ON zabbix.* TO 'zabbix'@'localhost';
SET GLOBAL log_bin_trust_function_creators = 1;
FLUSH PRIVILEGES;
EXIT;
```

Step 1.4: Import Zabbix Database Schema

Action

```
siem@ubuntu:~$ zcat /usr/share/zabbix-sql-scripts/mysql/server.sql.gz | mysql --default-character
# When prompted for password, type: SecureBank123!
```

```
# This takes 2-5 minutes. Wait for it to finish.
```

If You Get "Table 'role' already exists"

This means the schema was already imported from a previous attempt. You can skip this step and proceed to configuration.

Step 1.5: Configure Zabbix Server

Action

```
siem@ubuntu:~$ sudo nano /etc/zabbix/zabbix_server.conf
```

Find and set these values (uncomment if needed):

```
DBHost=localhost
DBName=zabbix
DBUser=zabbix
DBPassword=SecureBank123!
```

Save and exit: **Ctrl+O** , **Enter** , **Ctrl+X**

Step 1.6: Configure PHP Timezone

Action

```
siem@ubuntu:~$ sudo sed -i 's/;date.timezone =/date.timezone = Africa/Casablanca/' /etc/php/7.4/
siem@ubuntu:~$ grep "date.timezone" /etc/php/7.4/apache2/php.ini
```

Expected Output

```
date.timezone = Africa/Casablanca
```

Step 1.7: Create Zabbix Frontend Config

Action

```
siem@ubuntu:~$ sudo mkdir -p /etc/zabbix/web
siem@ubuntu:~$ sudo tee /etc/zabbix/web/zabbix.conf.php << 'EOF'
<?php
$DB['TYPE']      = 'MYSQL';
$DB['SERVER']    = 'localhost';
$DB['PORT']      = '0';
$DB['DATABASE']  = 'zabbix';
$DB['USER']      = 'zabbix';
$DB['PASSWORD']  = 'SecureBank123!';
$DB['SCHEMA']    = '';
$ZBX_SERVER      = 'localhost';
$ZBX_SERVER_PORT = '10051';
$ZBX_SERVER_NAME = 'SecureBank SOC';
$IMAGE_FORMAT_DEFAULT = IMAGE_FORMAT_PNG;
EOF
siem@ubuntu:~$ sudo chown www-data:www-data /etc/zabbix/web/zabbix.conf.php
siem@ubuntu:~$ sudo chmod 644 /etc/zabbix/web/zabbix.conf.php
siem@ubuntu:~$ sudo ln -sf /etc/zabbix/web/zabbix.conf.php /usr/share/zabbix/conf/zabbix.conf.php
```

Step 1.8: Unblock Admin Account (If Locked)

Action

```
siem@ubuntu:~$ sudo mysql -uroot -p
```

Inside MySQL:

```
USE zabbix;
UPDATE users SET attempt_failed = 0, attempt_clock = 0, passwd = MD5('zabbix') WHERE username = '
FLUSH PRIVILEGES;
EXIT;
```

Step 1.9: Start All Services

Action

```
siem@ubuntu:~$ sudo systemctl restart zabbix-server zabbix-agent apache2
siem@ubuntu:~$ sudo systemctl enable zabbix-server zabbix-agent apache2
siem@ubuntu:~$ sudo systemctl status zabbix-server
```

Step 1.10: Verify Database Connection

Action

```
siem@ubuntu:~$ php -r 'echo mysqli_connect("localhost", "zabbix", "SecureBank123!", "zabbix") ? "DB OK" : "DB FAIL";'
```

Expected Output

```
DB OK
```

Step 1.11: Access Zabbix Web Interface

Action

1. Open browser: `http://192.168.44.10/zabbix`
2. Username: `Admin`
3. Password: `zabbix`

If Setup Wizard Appears

Go through the setup wizard and enter:

- Database type: **MySQL**
- Database server: **localhost**
- Database port: **0** (or 3306)
- Database name: **zabbix**
- User: **zabbix**
- Password: **SecureBank123!**

Phase 1 Checklist

- ☐ Internet connectivity confirmed on SIEM
- ☐ Zabbix packages installed successfully
- ☐ MySQL database created and schema imported
- ☐ zabbix_server.conf has correct DB password
- ☐ PHP timezone configured
- ☐ Frontend config file created with correct password
- ☐ Admin account unblocked

- ☐ Services restarted and enabled
- ☐ Database connection test passes (DB OK)
- ☐ Login to <http://192.168.44.10/zabbix> works

Phase 2: Install Zabbix Agent on WEB Server

Host: WEB (192.168.44.129) | **User:** `vmweb` | **OS:** Ubuntu 18.04

Critical Note: Ubuntu 18.04 Compatibility

Ubuntu 18.04 has `libc6 2.27`. Zabbix Agent2 6.0 requires `libc6 >= 2.28` (Ubuntu 20.04+). **Use Zabbix Agent v1 instead.**

Step 2.1: Check Internet on WEB Server

Action

```
vmweb@ubuntu:~$ ping -c 3 google.com
vmweb@ubuntu:~$ ip addr show
```

Expected Result

IP should be `192.168.44.129`, ping to google.com should succeed.

Step 2.2: Add Zabbix Repository

Action

```
vmweb@ubuntu:~$ wget https://repo.zabbix.com/zabbix/6.0/ubuntu/pool/main/z/zabbix-release/zabbix-
vmweb@ubuntu:~$ sudo dpkg -i zabbix-release_*.deb
vmweb@ubuntu:~$ sudo apt update
```

Step 2.3: Install Zabbix Agent v1

Action

```
vmweb@ubuntu:~$ sudo apt install -y zabbix-agent
vmweb@ubuntu:~$ dpkg -l | grep zabbix
vmweb@ubuntu:~$ ls -la /etc/zabbix/
```

Expected Result

```
ii  zabbix-agent      1:6.0.xxx      amd64          Zabbix network monitoring solution - agent
drwxr-xr-x  2 root root   4096 ... /etc/zabbix/
-rw-r--r--  1 root root    ... /etc/zabbix/zabbix_agentd.conf
```

Step 2.4: Configure Agent

Action

```
vmweb@ubuntu:~$ sudo nano /etc/zabbix/zabbix_agentd.conf
```

Find and change these 3 lines:

Current Value	New Value
Server=127.0.0.1	Server=192.168.44.10
ServerActive=127.0.0.1	ServerActive=192.168.44.10
Hostname=Zabbix server	Hostname=web01-securebank

Save and exit: **Ctrl+O** , **Enter** , **Ctrl+X**

Quick Edit with sed (Alternative)

```
vmweb@ubuntu:~$ sudo sed -i 's/^Server=127.0.0.1/Server=192.168.44.10/' /etc/zabbix/zabbix_agentd
vmweb@ubuntu:~$ sudo sed -i 's/^ServerActive=127.0.0.1/ServerActive=192.168.44.10/' /etc/zabbix/z
vmweb@ubuntu:~$ sudo sed -i 's/^Hostname=Zabbix server/Hostname=web01-securebank/' /etc/zabbix/za
vmweb@ubuntu:~$ grep -E "^Server|^ServerActive|^Hostname" /etc/zabbix/zabbix_agentd.conf
```

Step 2.5: Start and Enable Agent

Action

```
vmweb@ubuntu:~$ sudo systemctl restart zabbix-agent
```

```
vmweb@ubuntu:~$ sudo systemctl enable zabbix-agent
vmweb@ubuntu:~$ sudo systemctl status zabbix-agent
```

Expected Output

```
• zabbix-agent.service - Zabbix Agent
  Loaded: loaded (...; enabled; ...)
  Active: active (running) since ...
  Main PID: ... (zabbix_agentd)
```

Step 2.6: Allow Firewall Port

Action

```
vmweb@ubuntu:~$ sudo ufw allow 10050/tcp
vmweb@ubuntu:~$ sudo ss -tlnp | grep 10050
```

Expected Output

```
LISTEN 0 4096 0.0.0.0:10050 0.0.0.0:* users:(("zabbix_agentd",...))
```

Phase 2 Checklist

- ☐ Internet connectivity confirmed on WEB
- ☐ Zabbix repository added
- ☐ Zabbix Agent v1 installed successfully
- ☐ Server=192.168.44.10 configured
- ☐ ServerActive=192.168.44.10 configured
- ☐ Hostname=web01-securebank configured
- ☐ Agent service running and enabled
- ☐ Port 10050 listening
- ☐ Firewall allows port 10050

Phase 3: Configure Host in Zabbix Dashboard

Location: Zabbix Web UI at <http://192.168.44.10/zabbix>

Step 3.1: Test Agent from SIEM

Action (on SIEM)

```
siem@ubuntu:~$ sudo apt install -y zabbix-get
siem@ubuntu:~$ zabbix_get -s 192.168.44.129 -k agent.ping
siem@ubuntu:~$ zabbix_get -s 192.168.44.129 -k system.hostname
```

Expected Output

```
1
ubuntu
```

1 means agent is responding. `ubuntu` is the hostname of WEB server.

Step 3.2: Create Host Group (If Missing)

Action

1. Login to Zabbix: <http://192.168.44.10/zabbix> (Admin / zabbix)
2. Go to **Configuration → Host groups**
3. Click **Create host group** (top right)
4. Group name: `Linux servers`
5. Click **Add**

Step 3.3: Create Host

Action

1. Go to **Configuration → Hosts**
2. Click **Create host** (top right)

Fill the form:

Field	Value
Host name	<code>web01-securebank</code> (must match agent config exactly!)
Visible name	<code>WEB Server - SecureBank</code>
Groups	Click Select → Choose Linux servers

Step 3.4: Add Agent Interface

Action

1. In the same host form, find **Interfaces** section
2. Click **Add** → Select **Agent**
3. IP address: 192.168.44.129
4. DNS name: *(leave empty)*
5. Port: 10050
6. Click **Add**

Step 3.5: Add Monitoring Template

Action

1. In the same host form, find **Templates** section
2. Click **Select**
3. Search: Linux by Zabbix agent
4. Select **Template OS Linux by Zabbix agent**
5. Click **Select** → Click **Add**

Critical: Template Must Be Added

Without a template, the host will show **GRAY** availability and collect no data. The template provides all monitoring items (CPU, memory, disk, network).

Step 3.6: Save Host

Action

1. Click **Add** at the bottom of the form
2. Wait 1-2 minutes for first data collection

Step 3.7: Remove Wrong Host (If Exists)

Action

1. Go to **Configuration** → **Hosts**
2. If you see "VM-WEB SecureBank" with IP 192.168.1.199

3. Select it (checkbox) → Click **Delete**

Phase 3 Checklist

- ☐ zabbix_get from SIEM returns 1
- ☐ zabbix_get from SIEM returns hostname "ubuntu"
- ☐ Host group "Linux servers" exists
- ☐ Host "web01-securebank" created
- ☐ Host IP set to 192.168.44.129
- ☐ Agent interface on port 10050
- ☐ Template "Template OS Linux by Zabbix agent" linked
- ☐ Wrong host "VM-WEB SecureBank" deleted (if existed)
- ☐ Host availability shows GREEN after 1-2 minutes

Phase 4: Verify Monitoring Data

Step 4.1: Check Host Availability

Action

1. Go to **Configuration → Hosts**
2. Look at **Availability** column for "WEB Server - SecureBank"

Expected Result

ZBX GREEN — Agent is connected and collecting data.

If still **GRAY** or **RED**, wait 2 more minutes and check troubleshooting section.

Step 4.2: View Latest Data

Action

1. Go to **Monitoring → Latest data**
2. Click **Hosts** dropdown
3. Select **WEB Server - SecureBank**

4. Click **Apply**

Expected Metrics

Metric	Example Value
CPU utilization	0.5% - 5%
Available memory	1.5 - 2.0 GB
Disk space (/) used	15% - 25%
Network ens33: Bits received	Active values
System load	0.01 - 0.50
Processes running	80 - 150

Step 4.3: Check Dashboard

Action

1. Go to **Monitoring → Dashboard**
2. Verify "Available" shows 2 hosts
3. Check that WEB Server metrics appear in widgets

Step 4.4: Add Dashboard Widget (Optional)

Action

1. Click **Edit dashboard** (top right)
2. Click **Add widget**
3. Type: **Graph**
4. Name: **WEB Server CPU**
5. Data set 1 → Host pattern: **Select** → "WEB Server - SecureBank"
6. Data set 1 → Item pattern: **Select** → "CPU utilization"
7. Click **Add**

Phase 4 Checklist

- ☐ Host availability is GREEN
- ☐ Latest data shows CPU metrics
- ☐ Latest data shows memory metrics

- ☐ Latest data shows disk metrics
- ☐ Latest data shows network metrics
- ☐ Dashboard shows 2 available hosts
- ☐ WEB Server data visible in dashboard

Phase 5: Configure Wazuh File Integrity Monitoring

Host: WEB (192.168.44.129)

Critical: Do NOT Use sed for XML Files

The `sed` command corrupts XML by inserting `<<` instead of `<`. Always use `tee` with heredoc for XML configuration files.

Step 5.1: Backup Current Config

Action

```
vmweb@ubuntu:~$ sudo cp /var/ossec/etc/ossec.conf /var/ossec/etc/ossec.conf.backup.$(date +%Y%m%d)
```

Step 5.2: View Current syscheck Section

Action

```
vmweb@ubuntu:~$ sudo grep -A20 "<syscheck>" /var/ossec/etc/ossec.conf
```

Expected Default Config

```
<syscheck>
  <disabled>no</disabled>
  <frequency>43200</frequency>
  <scan_on_start>yes</scan_on_start>
  <directories>/etc,/usr/bin,/usr/sbin</directories>
  <directories>/bin,/sbin,/boot</directories>
</syscheck>
```

Step 5.3: Add /var/www/html to syscheck (SAFE METHOD)

Action — Use tee with heredoc, NEVER sed

```
vmweb@ubuntu:~$ sudo bash -c 'cat > /tmp/syscheck_fix.xml << "XMLEOF"
<syscheck>
  <disabled>no</disabled>
  <frequency>300</frequency>
  <scan_on_start>yes</scan_on_start>
  <alert_new_files>yes</alert_new_files>
  <directories check_all="yes">/etc,/usr/bin,/usr/sbin</directories>
  <directories check_all="yes">/bin,/sbin,/boot</directories>
  <directories check_all="yes">/var/www/html</directories>
  <ignore>/etc/mtab</ignore>
  <ignore>/etc/hosts.deny</ignore>
  <ignore>/etc/mail/statistics</ignore>
</syscheck>
XMLEOF'

# Now use Python to safely replace the syscheck section
vmweb@ubuntu:~$ sudo python3 << 'PYEOF'
import re

with open('/var/ossec/etc/ossec.conf', 'r') as f:
    content = f.read()

with open('/tmp/syscheck_fix.xml', 'r') as f:
    new_syscheck = f.read().strip()

# Replace everything between <syscheck> and </syscheck>
pattern = r'<syscheck>.*?</syscheck>'
content = re.sub(pattern, new_syscheck, content, flags=re.DOTALL)

with open('/var/ossec/etc/ossec.conf', 'w') as f:
    f.write(content)

print("syscheck section updated successfully")
PYEOF
```

Why Python Instead of sed?

`sed` has escaping issues with XML tags. Python's `re.sub()` handles multi-line XML safely without corrupting angle brackets.

Step 5.4: Verify Config and Restart

Action

```
vmweb@ubuntu:~$ sudo grep -A15 "<syscheck>" /var/ossec/etc/ossec.conf
vmweb@ubuntu:~$ sudo systemctl restart wazuh-agent
```

```
vmweb@ubuntu:~$ sudo systemctl status wazuh-agent
```

Expected Output

- wazuh-agent.service - Wazuh agent
Active: active (running) ...
No errors about syscheck!

Step 5.5: Test File Integrity Monitoring

Action

```
vmweb@ubuntu:~$ echo "HACKED" | sudo tee /var/www/html/test_fim.html  
vmweb@ubuntu:~$ sudo rm /var/www/html/test_fim.html
```

Check Alerts on SIEM (after 5 minutes)

```
siem@ubuntu:~$ sudo grep -E "550|553|554" /var/ossec/logs/alerts/alerts.log | tail -10
```

Rule 550 = File added, 553 = File modified, 554 = File deleted

Phase 5 Checklist

- ☐ Config backed up before changes
- ☐ syscheck section verified with grep
- ☐ /var/www/html added to directories
- ☐ Frequency set to 300 seconds (5 min)
- ☐ alert_new_files set to yes
- ☐ No << or >> in XML (use Python to verify)
- ☐ Wazuh agent restarts without errors
- ☐ Test file creation triggers alert (after 5 min)

Troubleshooting Reference

T1: Zabbix Login Still Blocked

Symptom

"Incorrect user name or password or account is temporarily blocked" persists after fixes.

Fix

```
siem@ubuntu:~$ sudo mysql -uroot -p -e "USE zabbix; UPDATE users SET attempt_failed=0, attempt_clock=0"
siem@ubuntu:~$ sudo systemctl restart apache2
```

T2: Database Error "Access denied"

Symptom

"Access denied for user 'zabbix'@'localhost' (using password: YES)"

Fix

1. Check config file location: `/etc/zabbix/web/zabbix.conf.php`
2. Verify password is `SecureBank123!` not `zabbix123`
3. Remove any wrong config: `sudo rm -f /etc/zabbix/zabbix.conf.php`
4. Test DB: `php -r 'echo mysqli_connect("localhost","zabbix","SecureBank123!","zabbix")?"OK":"FAIL";'`

T3: Agent Shows GRAY Availability

Symptom

Host availability is GRAY instead of GREEN.

Fix Checklist

1. Verify agent is running on WEB: `sudo systemctl status zabbix-agent`
2. Verify port 10050 is listening: `sudo ss -tlnp | grep 10050`
3. Test from SIEM: `zabbix_get -s 192.168.44.129 -k agent.ping` → must return `1`
4. Check hostname matches exactly: Agent config = `web01-securebank`, Zabbix host name = `web01-securebank`
5. Verify template is linked: "Template OS Linux by Zabbix agent"
6. Check firewall on WEB: `sudo ufw allow 10050/tcp`
7. Wait 2 minutes after fixing, then refresh

T4: zabbix_get Returns "Check access restrictions"

Symptom

`zabbix_get` fails with connection or access error.

Fix

```
vmweb@ubuntu:~$ sudo nano /etc/zabbix/zabbix_agentd.conf
# Ensure Server line includes SIEM IP:
Server=192.168.44.10
vmweb@ubuntu:~$ sudo systemctl restart zabbix-agent
```

T5: Wazuh Agent Fails to Start

Symptom

Job for `wazuh-agent.service` failed with "Invalid element in the configuration: 'syscheck'"

Fix

1. Restore from backup: `sudo cp /var/ossec/etc/ossec.conf.backup.YYYYMMDD /var/ossec/etc/ossec.conf`
2. If no backup, use the Python script from Step 5.3 to rebuild syscheck
3. Never use `sed` on XML files — use `tee` + heredoc or Python
4. Verify with: `sudo grep -c "<<" /var/ossec/etc/ossec.conf` → should return `0`

T6: Zabbix Agent2 Won't Install on Ubuntu 18.04

Symptom

`zabbix-agent2` : Depends: `libc6 (>= 2.28)` but `2.27-3ubuntu1.5` is to be installed

Fix

Use Zabbix Agent v1 instead. It is fully compatible:

```
vmweb@ubuntu:~$ sudo apt install -y zabbix-agent
# Config file: /etc/zabbix/zabbix_agentd.conf
# Service name: zabbix-agent
```

Quick Command Reference

Task	Command	Host
Restart Zabbix Server	<code>sudo systemctl restart zabbix-server</code>	SIEM
Restart Zabbix		

Agent	<code>sudo systemctl restart zabbix-agent</code>	WEB
Restart Apache	<code>sudo systemctl restart apache2</code>	SIEM
Restart Wazuh Agent	<code>sudo systemctl restart wazuh-agent</code>	WEB
Test agent from SIEM	<code>zabbix_get -s 192.168.44.129 -k agent.ping</code>	SIEM
Check agent status	<code>sudo systemctl status zabbix-agent</code>	WEB
Check agent port	<code>sudo ss -tlnp grep 10050</code>	WEB
View Zabbix server log	<code>sudo tail -20 /var/log/zabbix/zabbix_server.log</code>	SIEM
View Apache errors	<code>sudo tail -20 /var/log/apache2/error.log</code>	SIEM
View Wazuh agent log	<code>sudo tail -20 /var/log/zabbix/zabbix_agentd.log</code>	WEB
Check Wazuh alerts	<code>sudo tail -20 /var/ossec/logs/alerts/alerts.log</code>	SIEM
DB connection test	<code>php -r 'echo mysqli_connect("localhost","zabbix","SecureBank123!","zabbix")?"OK":"FAIL";'</code>	SIEM

Implementation Guide Complete

SecureBank SOC Monitoring Infrastructure

Follow phases in order: 1 → 2 → 3 → 4 → 5

Use Troubleshooting section if any step fails